Experiment 3

AIM: Develop Expense Tracking Module with Exception Handling.

Source: Program.cs

Code:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Practical_3
{
    class Expense
    {
        public int Expid;
        public string Category;
        public double Amount;
        public string Payment_Method;
        public DateTime Date;


        public void addExpense()
        {
            Console.Write("Enter Expense ID:");
            Expid = Convert.ToInt32(Console.ReadLine());


            Console.Write("Enter Category:");
            Category = Console.ReadLine();

            Console.Write("Enter Amount:");
            Amount = Convert.ToDouble(Console.ReadLine());

            try
            {
                if (Amount < 0)
                {
                    throw new ArgumentException("Expense cannot be negative.");
                }
            }
            catch (ArgumentException ex)
            {
                Console.WriteLine(ex.Message);
                Console.Write("Enter Amount:");
                Amount = Convert.ToDouble(Console.ReadLine());
            }

            Console.Write("Enter Payment Method:");
            Payment_Method = Console.ReadLine();

            Console.Write("Enter Date (DD-MM-YYYY):");
            Date = DateTime.Parse(Console.ReadLine());
        }
```

```
    public void displayExpense()
    {
        Console.WriteLine("Expense ID: " + Expid);
        Console.WriteLine("Category: " + Category);
        Console.WriteLine("Amount: " + Amount);
        Console.WriteLine("Payment Method: " + Payment_Method);
        Console.WriteLine("Date: " + Date.ToString("yyyy-MM-dd"));
    }
}

internal class Program
{
    static void Main(string[] args)
    {
        List<Expense> expenses = new List<Expense>();
        int i = 0;
        do
        {
            Console.WriteLine("=============================================");
            Console.WriteLine("         Expense Management System");
            Console.WriteLine("=============================================");
            Console.WriteLine("1. Add Expense");
            Console.WriteLine("2. Display Expenses");
            Console.WriteLine("3. Total Expense Amount");
            Console.WriteLine("4. Exit");
            Console.Write("Enter your choice: ");

            try
            {
                i = Convert.ToInt32(Console.ReadLine());
                Console.WriteLine();
            }
            catch (Exception ex)
            {
                Console.WriteLine();
                Console.WriteLine("-----------------------------------------------");
                Console.WriteLine("Enter Valid Integer Choice");
                Console.WriteLine("-----------------------------------------------");
                Console.WriteLine();
            }
            Expense e = new Expense();
            if (i == 1)
            {
                e.addExpense();
                expenses.Add(e);
                Console.WriteLine("-----------------------------------------------");
                Console.WriteLine("Expense Added Successfully");
                Console.WriteLine("-----------------------------------------------");
                Console.WriteLine();
                Console.WriteLine();

            }
            else if (i == 2)
            {
                foreach (Expense exp in expenses)
                {
                    exp.displayExpense();
```

```
                    Console.WriteLine("-----------------------------------------------");
                    Console.WriteLine();
                }
            }
            else if (i == 3)
            {
                double sum = 0;
                foreach (Expense exp in expenses)
                {
                    sum += exp.Amount;

                }
                Console.WriteLine();
                Console.WriteLine("-----------------------------------------------");
                Console.WriteLine("Total Expense Amount: " + sum);
                Console.WriteLine("-----------------------------------------------");
                Console.WriteLine();

            }
            else if (i == 4)
            {
                Console.WriteLine();
                Console.WriteLine("-----------------------------------------------");
                Console.WriteLine("Exiting the Program");
                Console.WriteLine("-----------------------------------------------");
                Console.WriteLine();
            }
        } while (i < 4);
    }
  }
}
```

Output:

```
E:\Study\Marwadi\Sem-5\Ne
===========================================
        Expense Management System
===========================================
1. Add Expense
2. Display Expenses
3. Total Expense Amount
4. Exit
Enter your choice: 1

Enter Expense ID:1
Enter Category:Food
Enter Amount:1500
Enter Payment Method:UPI
Enter Date (DD-MM-YYYY):15/08/2026
-----------------------------------------------
Expense Added Successfully
-----------------------------------------------


===========================================
        Expense Management System
===========================================
1. Add Expense
2. Display Expenses
3. Total Expense Amount
4. Exit
Enter your choice: 1

Enter Expense ID:2
Enter Category:Car
Enter Amount:15000
Enter Payment Method:CASH
Enter Date (DD-MM-YYYY):20/09/2025
-----------------------------------------------
Expense Added Successfully
-----------------------------------------------
```

```
==========================================
         Expense Management System
==========================================
1. Add Expense
2. Display Expenses
3. Total Expense Amount
4. Exit
Enter your choice: 2

Expense ID: 1
Category: Food
Amount: 1500
Payment Method: UPI
Date: 2026-08-15
------------------------------------------

Expense ID: 2
Category: Car
Amount: 15000
Payment Method: CASH
Date: 2025-09-20
------------------------------------------

==========================================
         Expense Management System
==========================================
1. Add Expense
2. Display Expenses
3. Total Expense Amount
4. Exit
Enter your choice:
```

```
==============================================
          Expense Management System
==============================================
1. Add Expense
2. Display Expenses
3. Total Expense Amount
4. Exit
Enter your choice: 3


----------------------------------------------
Total Expense Amount: 16500
----------------------------------------------
```

```
==============================================
          Expense Management System
==============================================
1. Add Expense
2. Display Expenses
3. Total Expense Amount
4. Exit
Enter your choice: 4


----------------------------------------------
Exiting the Program
----------------------------------------------
```

Github Url: https://github.com/jd-thakrar/Uni-DotNet-Work/tree/master/Practical-3